

# NOVITA FITRIYANTI PULUNGAN

Mechanical Engineering Graduate | Production Planning, Inventory & Procurement Support | Excel  
Bogor, West Java (open to relocation)  
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## PROFESSIONAL SUMMARY

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Mechanical Engineering graduate who turned raw sensor data into usable results in two team-based experimental research projects, processing thousands of data rows per run in Excel at BRIN. Prepared the budget plan and a 55-transaction financial report as treasurer of a 66-person student event, and handled cashier, opening, and closing duties at an independent coffee shop, including running it alone. Seeking production planning (PPIC), inventory control, warehouse/logistics administration, or procurement roles in manufacturing.

## EDUCATION

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**Udayana University** 2020 – 2024  
*Bachelor of Engineering in Mechanical Engineering* | Bali, Indonesia  
GPA 3.84/4.00

**Thesis:** Experimental Study of Two-Phase Flow Characteristics of Non-Newtonian Fluid Xanthan Gum (XG) 0.1% and Nitrogen in Horizontal Microchannel with Sudden Expansion

**Relevant coursework:** Production Processes I & II, Quality Control, Industrial Metrology, Engineering Economics, Statistics

## TECHNICAL & RESEARCH EXPERIENCE

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**Undergraduate Thesis Research (3-person team)** Sep 2023 – Jul 2024  
*Pressure Measurement & Data Processing*

Fluid Mechanics Laboratory, Universitas Gadjah Mada & Udayana University | Yogyakarta & Bali

- Processed raw signals from 5 differential pressure transducers across 28 flow conditions for 2 liquids, converting sensor voltage to kPa with per-sensor calibration equations in Excel to produce pressure-drop results.
- Calibrated the pressure transducers with the team using a liquid-column method (10–50 cm steps) and derived a linear calibration equation for each sensor in Excel.
- Compared pressure drop across flow conditions and liquids, showing that the 0.1% xanthan gum solution consistently produced a higher pressure drop than water.
- Co-drafted, with two fellow students, the Kampus Merdeka Research Grant proposal (Udayana University) that funded the team's research at Universitas Gadjah Mada; the grant was held by the supervising lecturer.

**Research Intern – Thermohydraulics Laboratory (PRTRN)** Sep – Dec 2022  
*National Research and Innovation Agency (BRIN)* | Serpong, Banten

- Processed Excel exports from an NI cDAQ data acquisition system (thousands of rows per run) on the thermal behavior of nanobubbles during heating, 1-hour steady state, and cooling at 70, 80, and 90°C, as part of a 4-person team.
- Supported thermocouple calibration before data collection and monitored temperatures during experimental runs.
- Plotted temperature data and heat transfer coefficient results (calculated with the supervisor's model) in Origin.
- Wrote and maintained the team's group reports and presented findings to the BRIN supervisor and the visiting Udayana University supervisor.

## ORGANIZATIONAL EXPERIENCE

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### Mechanical Engineering Student Association (HMM)

Jan – Dec 2022

*Staff, Student Resource Development Division (PSDM) | Udayana University | Bali*

- *Treasurer, Mechanical Talk 2022* (3-day orientation, 66-person committee): prepared the IDR 5,380,000 budget plan and a 55-transaction financial report, recording IDR 2,309,000 in income against IDR 2,286,500 in expenses.
- Coordinated the fundraising and catering teams for Mechanical Talk 2022; snack sales raised IDR 2,220,000, covering about 97% of event expenses.
- *Secretary, Student Creativity Program (PKM) Training Month 2022* (approx. 300 participants in 60 groups): drafted the IDR 7 million funding request (IDR 2,100,000 disbursed by the study program), recorded 8 sets of meeting minutes, and compiled the accountability report.
- *Public Relations Coordinator, Scientific Writing Class 2022*: led a 3-person team that sent official letters to 20 partner institutions and served as the event's contact person.

## OTHER WORK EXPERIENCE

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### Freelance Sales Booth Staff

Nov – Dec 2024

*Bazaar Events (multiple brands) | Bali*

- Set up product displays, served and sold to customers, handled cash transactions (including basic English with international customers), and closed the booth on 12-hour event days.
- Referred by brand owners to other brands for weekend booth shifts.

### Part-time Barista & Store Operations

Mar – Oct 2024

*Independent Coffee Shop | Bali*

- Opened and closed the shop, prepared coffee and simple drinks, took orders, and handled the cash register, with the owner as the only other staff member.
- Ran the shop alone when the owner was away, while completing an undergraduate thesis.

### Helpdesk Staff

May 2024

*10th World Water Forum | Bali*

- Provided session, venue, and logistics information to 70+ international delegates.

## SKILLS

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- Excel: SUMIFS, XLOOKUP, IF logic, PivotTables, dashboards
- Origin: experimental data processing and plotting
- Microsoft Word, PowerPoint, Google Workspace
- Languages: Indonesian (native); English – intermediate (written, basic conversation)